radioactive metal will and won't emit radiation.

Thanks to these punk-rock particles, the Geiger counter will and won't register radiation, which means the hammer will and won't smash the vial of cyanide—so the cat will be both dead and alive. And this, Schrödinger pointed out, is a serious problem. Maybe an atom can travel down two paths at once, but a cat certainly can't be both dead and alive. When we open the box, the cat will be either dead or alive, and it stands to reason that the cat must have been one or the other the moment before we opened the box.

Yet many of Schrödinger's contemporaries piled on, denying exactly that point. Some claimed that the cat was in a state of dead-and-alive until the moment the box was opened, when the cat was somehow forced into "aliveness" or "deadness" through the action of looking inside the box. Others believed that talking about what was going on inside the box before it was opened was meaningless, because the interior of the unopened box was unobservable by definition, and only observable, measurable things have meaning. To them, worrying about unobservable things was pointless, like asking whether a tree that falls in the forest makes a sound when nobody's around to hear it.

Schrödinger's concerns about his cat weren't allayed by these arguments. He thought that his colleagues had missed the point: quantum physics lacked an important component, a story about how it lined up with the things in the world. How does a phenomenal number of atoms, governed by quantum physics, give rise to the world we see around us? What is real, at the most fundamental level, and how does it work? Yet Schrödinger's opponents carried the day, and his concerns about what was actually happening in the quantum world were dismissed. The rest of physics simply moved on.

Schrödinger was in a minority, but he wasn't alone. Albert Einstein also wanted to understand what was really happening in the quantum world. He debated Niels Bohr, the great Danish physicist, over the nature of quantum physics and reality. The Einstein-Bohr debates have entered into the lore of physics itself, and the usual conclusion is that Bohr won, that Einstein's and